Lecture 5: Omitted Variable Bias

In the last lecture, we discussed the assumptions needed for consistency and unbiasedness of ordinary least squares. Of these assumptions, the one we focus on the most is the orthogonality condition, which states that the error term is uncorrelated with the regressor. In this lecture and the following lecture, we will focus on different situations in which the orthogonality condition may be violated. Today we will focus on omitted variable bias: a situation in which a variable that is not included as a regressor causes a failure of orthogonality.

The Orthogonality Condition and Omitted Variable Bias

To keep track of notation, let us again recall the (causal) linear model:

$$Y_i = \alpha + \beta_1 X_{1,i} + ... + \beta_K X_{K,i} + \varepsilon_i$$

where Y_i is the outcome, α is the intercept, β_k is the causal effect of the k-th regressor, $X_{k,i}$, and ε_i is the error term, which reflects everything else that affects the outcome and was not included as a regressor.

Last time, we showed that in order for OLS to be consistent, it must be the case that the orthogonality condition holds. This condition stated that the error has mean zero and is uncorrelated with the regressors:

$$\mathbb{E}\left[\varepsilon_i\right] = 0$$
$$\text{Cov}\left(X_{k,i}, \varepsilon_i\right) = 0 \text{ for all } k$$

For unbiasedness, we needed the somewhat stronger zero conditional mean assumption, $\mathbb{E}\left[\varepsilon_i \mid X_{1,i}, ..., X_{K,i}\right] = 0$. However, since this zero conditional mean assumption implies the orthogonality condition, it is also true that orthogonality is *necessary* for OLS to be unbiased (it just isn't quite sufficient!).

Today, we will study what happens when we exclude important variables from our regression. To start, let's focus on the case where we run a regression with a single regressor, $X_{1,i}$, but the true causal model also has $X_{2,i}$. So, suppose the true causal model is:

$$Y_i = \alpha + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \varepsilon_i$$

and that the orthogonality condition holds for this true model. However, the econometrician instead estimates the regression model:

$$Y_i = \tilde{\alpha} + \tilde{\beta}_1 X_{1,i} + \tilde{\varepsilon}_i$$

Suppose that we have a large sample, and our other regression assumptions (no perfect collinearity + IID sampling) are satisfied, so our estimate of $\tilde{\beta}_1$ will converge. What will our estimate of $\tilde{\beta}_1$ converge to? Using our formula for the slope in a single regressor model, we have:

$$\begin{aligned}
\tilde{\beta}_1 &= \frac{\text{Cov}\left(X_{1,i}, Y_i\right)}{\text{Var}\left(X_{1,i}\right)} = \frac{\text{Cov}\left(X_{1,i}, \alpha + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \varepsilon_i\right)}{\text{Var}\left(X_{1,i}\right)} \\
&= \frac{\text{Cov}\left(X_{1,i}, \beta_1 X_{1,i}\right) + \text{Cov}\left(X_{1,i}, \beta_2 X_{2,i}\right) + \text{Cov}\left(X_{1,i}, \varepsilon_i\right)}{\text{Var}\left(X_{1i}\right)} \\
&= \frac{\beta_1 \cdot \text{Var}\left(X_{1,i}\right) + \beta_2 \cdot \text{Cov}\left(X_{1,i}, X_{2,i}\right) + 0}{\text{Var}\left(X_{1,i}\right)} \\
&= \beta_1 + \beta_2 \cdot \underbrace{\frac{\text{Cov}\left(X_{1i}, X_{2,i}\right)}{\text{Var}\left(X_{1,i}\right)}}_{\text{Regression of } X_{2,i} \text{ on } X_{1,i}} \\
&= \beta_1 + \underbrace{\beta_2 \cdot \tilde{\delta}_1}_{\text{Omitted Variable Bias}}
\end{aligned}$$

where $\tilde{\delta}_1$ is the coefficient from a regression of $X_{2,i}$ on $X_{1,i}$. Thus, in the case with one included regressor and one omitted regressor, the "naive" coefficient $\tilde{\beta}_1$ will reflect the true coefficient, β_1, plus an omitted variable bias, $\beta_2 \cdot \frac{\text{Cov}(X_{1,i}, X_{2,i})}{\text{Var}(X_{1,i})}$. We can see that $\frac{\text{Cov}(X_{1,i}, X_{2,i})}{\text{Var}(X_{1,i})}$ is the coefficient we would get from regressing $X_{2,i}$ on $X_{1,i}$. We can remember this by saying that the naive coefficient $\tilde{\beta}_1$ is equal to the true causal effect of $X_{1,i}$ on the outcome, plus the spurious "effect" of $X_{1,i}$ through its effect on $X_{2,i}$. Note however that it need not actually be the case that $X_{1,i}$ "causes" $X_{2,i}$: just the fact that they are correlated is enough.

This omitted variable bias formula reveals to us when it is the case that we need to worry about an omitted variable. The omitted variable bias is the product of β_2 and $\tilde{\delta}_1$, so it is only non-zero if both of those coefficients are non-zero. In order for the omitted regressor $X_{2,i}$ to create an omitted variable bias, two things must be true:

- **The omitted variable (causally) affects the outcome:** $\beta_2 \neq 0$
- **The omitted variable is correlated with the included regressor:** $\tilde{\delta}_1 = \frac{\text{Cov}(X_{1,i}, X_{2,i})}{\text{Var}(X_{1,i})} \neq 0$

If either of the above conditions does not hold, then we don't need to worry about omitted variable bias.

We can also use the omitted variable bias formula to think about the sign and magnitude of potential omitted variable biases. If an omitted variable has a positive effect on the outcome and a positive correlation with the included regressor, then it will create a positive bias in the estimate: the estimated effect $\tilde{\beta}_1$ will be larger than the true effect β_1. If one is positive and the other is negative, we will have a negative bias: the naive estimate will be too small relative to the true causal effect. If both are negative, then we will have a positive bias.

A sophisticated researcher can also use the omitted variable bias formula to think about the magnitude of a potential omitted variable bias. For example, if we know that a regression of $X_{2,i}$ on $X_{1,i}$ would yield a very small coefficient, then we can say that the omitted variable bias is likely to be small (unless the causal effect β_2 is very large!).

A Short Example

An empirical regularity across data sets is that taller people earn more money. This effect holds true even when you focus on within-gender differences. In a sample of British 30-year-olds who were born in 1970, Case and Paxson (2008) run the following regression:[1]

$$\log \text{Hourly Earnings}_i = \alpha + \beta \cdot \text{Height in Inches}_i + \varepsilon_i$$

where log is the natural logarithm. In their sample of British men, they estimate $\hat{\beta} = 0.010$, meaning that on average, a man who is one inch taller would have roughly 1% higher earnings. For a (hypothetical) 5 foot 4 inch man, this would mean that growing to the average height of 5 foot 10 inches would raise his earnings by roughly 6%. For women, they estimate a coefficient of $\hat{\beta} = 0.015$.

A causal interpretation of this regression would imply that simply being taller, without changing anything else, would increase a person's income. Since height does not actually affect most worker's productivity in a modern economy, one explanation could be conscious or unconscious discrimination against short people.

Instead, Case and Paxson argue that the relationship reflects omitted variable bias. For example, taller people also tend to have higher measured IQs. Case and Paxson run a new regression controlling for IQ measured at age 5 and 10:

$$\log \text{Hourly Earnings}_i = \alpha + \beta_1 \cdot \text{Height in Inches}_i + \beta_2 \text{IQ at Age 5}_i + \beta_3 \text{IQ at Age 10}_i + \varepsilon_i$$

They now estimate $\hat{\beta}_1 = 0.004$ for men and $\hat{\beta}_1 = 0.006$ for women, cutting the previous estimates by more than half. Adding in additional controls (e.g. for parent's socioeconomic background and conditions in childhood) cuts the coefficients to be near-zero and statistically insignificant.

[1] All of Case and Paxson's regressions also control for ethnicity, but I have omitted that for presentational simplicity.

In this example, there was omitted variable bias because height and IQ are correlated, and because IQ has a causal effect on income. Since the correlation between height and IQ is positive, and the causal effect of IQ on income is positive, the bias is positive: the true causal effect of height on income is smaller than the estimate we would get from the "short regression" (the regression that is missing a variable). Note that omitted variable bias does *not* require the relationship between the included and omitted regressor to be causal. Being taller does not cause people to be more intelligent, instead, poor health in utero or in childhood can negatively affect both height and IQ. But just the correlation between the regressor and the omitted variable, coupled with the fact that the omitted variable affects income, is sufficient to create omitted variable bias.

Exercise 1. Consider the linear model $Y_i = \alpha + \beta \cdot X_i + \varepsilon_i$. For each of the following examples, think of (a) some variables that are likely to *causally* affect the outcome, and (b) some variables that are likely to be *correlated* with the regressor.

- **Fiscal Multipliers:** $Y_t =$ Output Growth, $X_t =$ Government Spending Growth
- **Returns to Job Training:** $Y_i =$ Income, $X_i =$ Participation in Job Retraining Program
- **Returns to Capital:** $Y_i =$ Quarterly Output, $X_i =$ Firm's Capital Stock
- **Healthcare:** $Y_i =$ Indicator for Mortality, $X_i =$ Went to Hospital in Last Month
- **Income-Mortality:** $Y_i =$ Age at Death, $X_i =$ Income at Age 40
- **Crime:** $Y_i =$ Neighborhood Crime Rate, $X_i =$ Police Officers per Capita

Exercise 2. For each example in the exercise above, are there any variables that satisfy both (a) and (b)? What is the likely sign of the omitted variable bias coming from that omitted variable?

Bonus Material: Omitted Variable Bias with Many Regressors

What if we instead have many regressors? The omitted variable bias formula can be extended to the many regressors case.

One Regressor and Many Omitted Variables

First, suppose that we run a regression with one regressor, $X_{1,i}$ but there are many omitted regressors, $X_{2,i} \ldots X_{K,i}$. In that case, the omitted variable bias formula is simply a sum of each individual omitted variable bias:

$$\begin{aligned}
\tilde{\beta}_1 &= \frac{\text{Cov}(X_{1,i}, Y_i)}{\text{Var}(X_{1,i})} \\
&= \frac{\text{Cov}(X_{1,i}, \alpha + \beta_1 X_{1,i} + \ldots + \beta_K X_{K,i} + \varepsilon_i)}{\text{Var}(X_{1,i})} \\
&= \beta_1 + \beta_2 \cdot \frac{\text{Cov}(X_{1,i}, X_{2,i})}{\text{Var}(X_{1,i})} + \ldots + \beta_K \cdot \frac{\text{Cov}(X_{1,i}, X_{K,i})}{\text{Var}(X_{1,i})}
\end{aligned}$$

Many Regressors and One Omitted Variable

The other case is where we include regressors $X_{1,i}$ through $X_{K,i}$, but we omit $X_{K+1,i}$. How does omitted variable bias affect our estimate $\tilde{\beta}_1$?

To answer this question, it will be useful to first have what is called the Frisch-Waugh-Lovell (FWL) Theorem, which helps us to better understand multiple regression.

Theorem (Frisch-Waugh-Lovell Theorem)**.** *Consider the linear model*

$$Y_i = \alpha + \beta_1 X_{1,i} + ... + \beta_K X_{K,i} + \varepsilon_i$$

Let $\tilde{X}_{1,i}$ *denote the "residualized"* $X_{1,i}$*, defined as the residual of a regression of* $X_{1,i}$ *on* $X_{2,i}...X_{K,i}$*. Define* $\tilde{Y}_i$ *similarly, as the residual from a regression of* Y_i *on* $X_{2,i}...X_{K,i}$*. Then, the OLS estimate* $\hat{\beta}_1$ *will be given by* $\hat{\beta}_1 = \frac{\widehat{Cov}(\tilde{X}_{1,i}, Y_i)}{\widehat{Var}(\tilde{X}_{1,i})} = \frac{\widehat{Cov}(\tilde{X}_{1,i}, \tilde{Y}_i)}{\widehat{Var}(\tilde{X}_{1,i})}$.

This theorem tells us that we can think of the coefficient $\hat{\beta}_1$ in a regression with multiple regressors as being equivalent to the coefficient from a regression with a single regressor, but where we have first residualized the regressor $X_{1,i}$ on all of the other regressors (we can also residualize the outcome in the same way, although it actually doesn't matter for the coefficient estimate if we do this or not).

This theorem is very useful in general, but we will now specifically use it to think about omitted variable bias with multiple regressors. If we we include regressors $X_{1,i}$ through $X_{K,i}$, but we omit $X_{K+1,i}$, then we can use the FWL Theorem to write the (naive) estimate $\tilde{\beta}_1$:

$$\begin{aligned}
\tilde{\beta}_1 &= \frac{\text{Cov}\left(\tilde{X}_{1,i}, Y_i\right)}{\text{Var}\left(\tilde{X}_{1,i}\right)} \\
&= \frac{\text{Cov}\left(\tilde{X}_{1,i}, \alpha + \beta_1 X_{1,i} + ... + \beta_K X_{K,i} + \beta_{K+1} X_{K+1,i} + \varepsilon_i\right)}{\text{Var}\left(\tilde{X}_{1,i}\right)} \\
&= \frac{\text{Cov}\left(\tilde{X}_{1,i}, \beta_1 X_{1,i} + \beta_{K+1} X_{K+1,i}\right)}{\text{Var}\left(\tilde{X}_{1,i}\right)} \\
&= \beta_1 + \beta_{K+1} \underbrace{\frac{\text{Cov}\left(\tilde{X}_{1,i}, X_{K+1,i}\right)}{\text{Var}\left(\tilde{X}_{1,i}\right)}}_{\text{Regression of } X_{K+1,i} \text{ on } X_{1,i}\text{, controlling for } X_{2,i}...X_{K,i}}
\end{aligned}$$

Thus, the omitted variable bias formula is very similar to before, the difference now being that instead of having the regression coefficient of the omitted variable on the included regressor, we now have the regression of the omitted variable on the included regressor, controlling for the other included regressors.